

“STAB X”

Software for optical stabilization



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Description



Works on Raspberry Pi Zero and Raspberry Pi 5 (tested up to **1500** meters altitude). Currently implementing on a custom board and Ukrainian thermal imaging cameras.



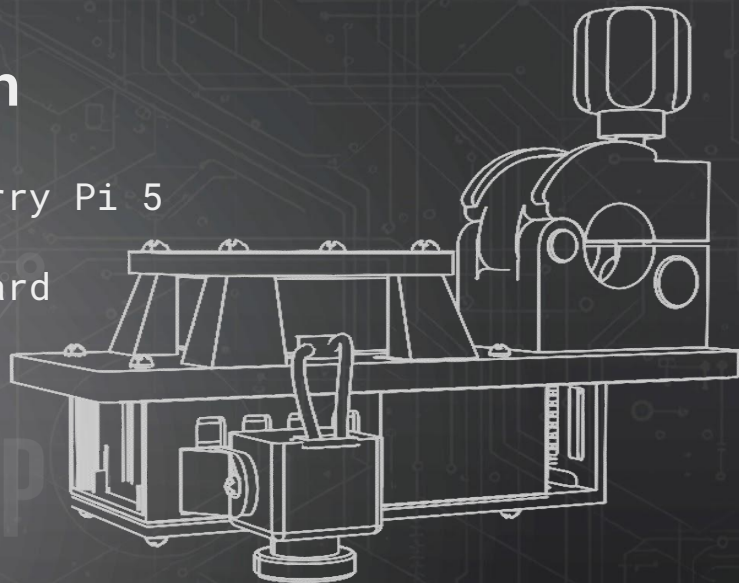
Night mode support



Stabilization activates from the moment of takeoff. Control is simpler than a Mavic 3. An untrained pilot can easily complete the mission.



Copy protection – tied to a specific RPI and SD card.



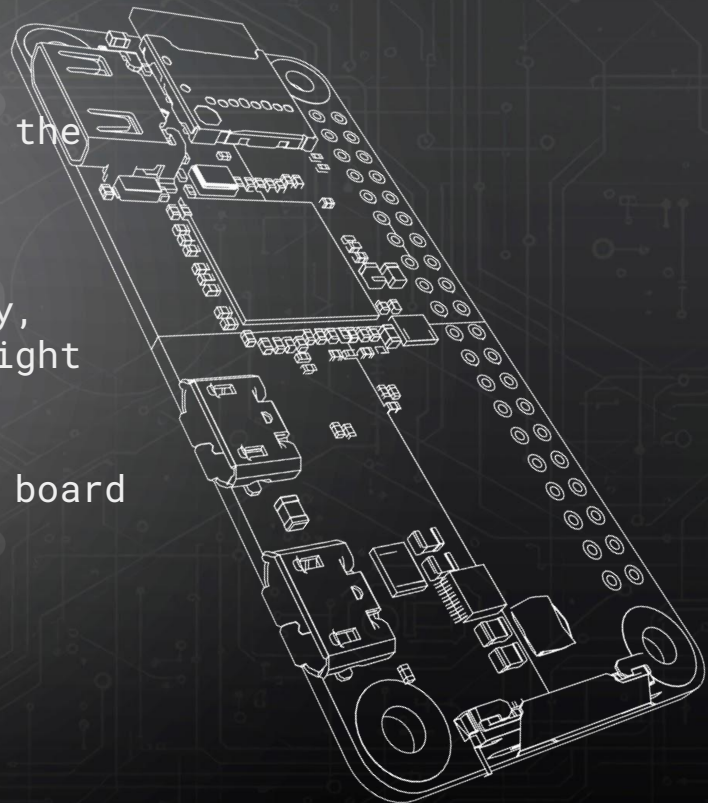
Technological Features

Simple Power and Integration

Raspberry Pi Zero receives power directly from the flight controller (the simplest, most reliable solution).

Raspberry Pi 5 requires a separate power supply, making it more difficult to solder into the flight controller.

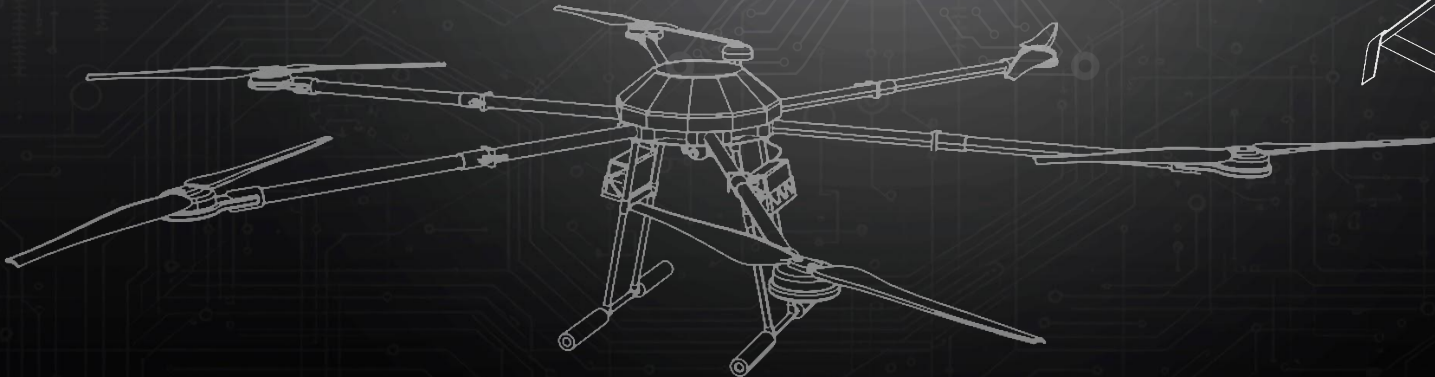
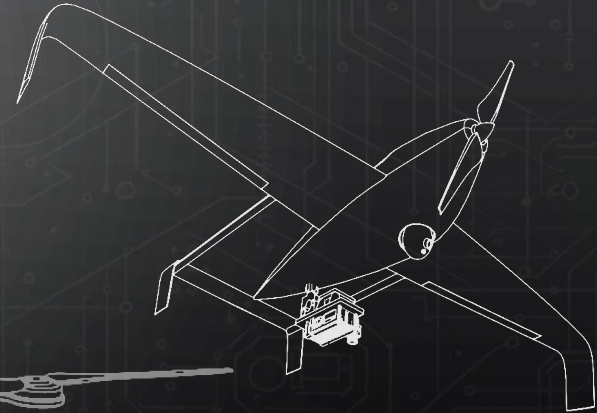
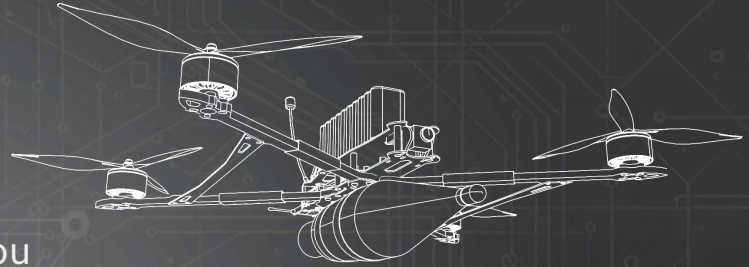
In the future - a custom, smallest form-factor board that receives power directly from the flight controller.



Stability in the Air

The autopilot does not take over control—you can still pilot the drone even during active stabilization.

Suitable for reconnaissance drones, relay drones, and bomber drones (ensures high positioning accuracy).

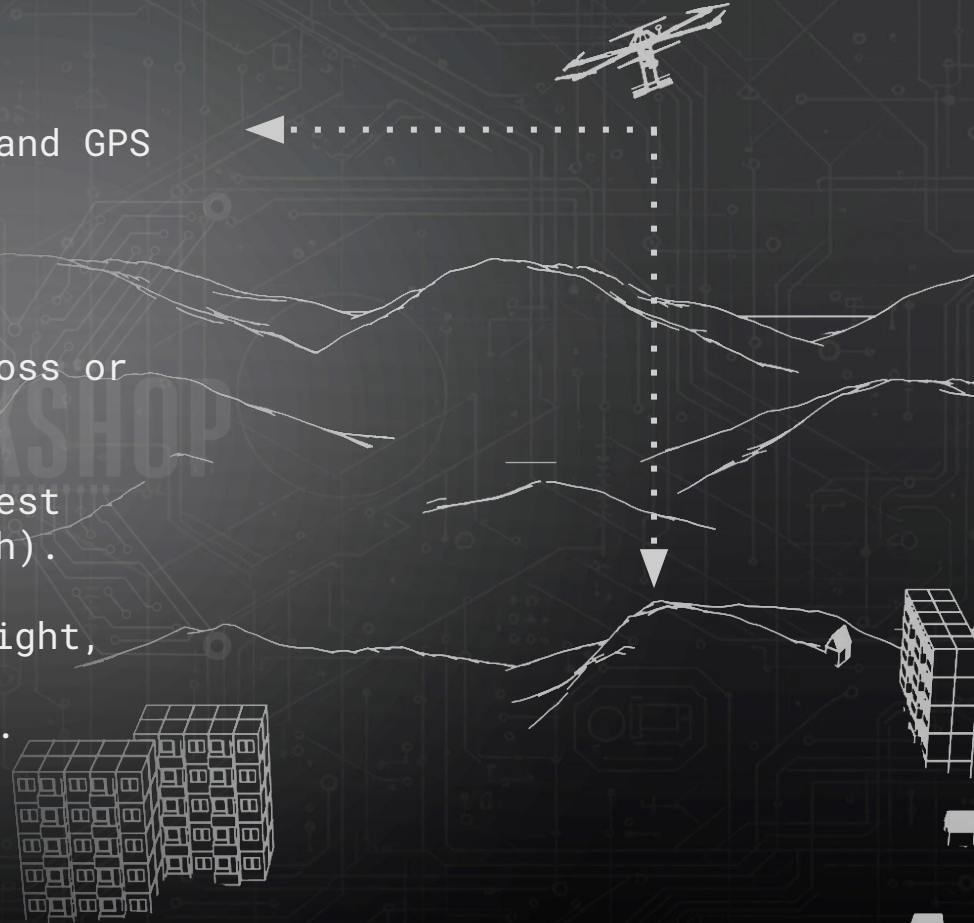


Maintaining Azimuth Without Compass and GPS
Drift no more than 1
1° per 5 minutes.

Return-to-Home (RTL) Function
Automatic return in case of signal loss or
suppression.

Return to the operator via the shortest
trajectory (not along the flight path).

Return error ~200 m over a 7-8 km flight,
depends on the terrain
(being refined to increase accuracy).



Positioning Accuracy



Altitude	Deviation (Horizontal)	Deviation (Vertical)
50 m	~1 m	
100-200 m	~2 m	
500 m	3-5 m*	
1000 m	10-15 m*	10 m at a wind speed of 10 m/s

* Operation at these altitudes is only possible with custom firmware, with corrected navigation limitations that make stabilization at heights of 400+ meters smoother.

** Video of operation from the front line is provided upon request.

Protection and Safety



Activation via Bluetooth Key

Without the activator key, stabilization won't work (a safeguard in case the drone is lost).

The ability to add multiple keys to a single license. Activation control - tracking key usage on computers.



Copy Protection

After activation, the software only works on a specific RPi (Raspberry Pi) and SD card. The image cannot be copied-the program is encrypted, and the key remains with the pilot.



Easy Integration

Connection with just 4 wires

Changing a few parameters in Ardupilot for correct operation.

Thank you for your attention!

